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(54) **DISPENSER ASSEMBLY INCLUDING AN
OPEN CHANNEL LINE PURGE**

(52) **U.S. Cl.**

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ABSTRACT

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A method of operating a dispenser assembly of an appliance includes receiving a signal to dispense a predetermined volume of a fluid from at least one fluid storage vessel via a fluid diverter; selecting a first fluid inlet channel in response to receiving the signal, the first fluid inlet channel being fluidly connected with the at least one fluid storage vessel; initiating a pump to pump the predetermined volume of the fluid from the at least one fluid storage vessel after selecting the first fluid inlet channel; and selecting a second fluid inlet channel of the plurality of fluid inlet channels after pumping the predetermined volume of the fluid, the second fluid inlet channel being an open-air inlet channel.

(21) Appl. No.: **18/585,200**

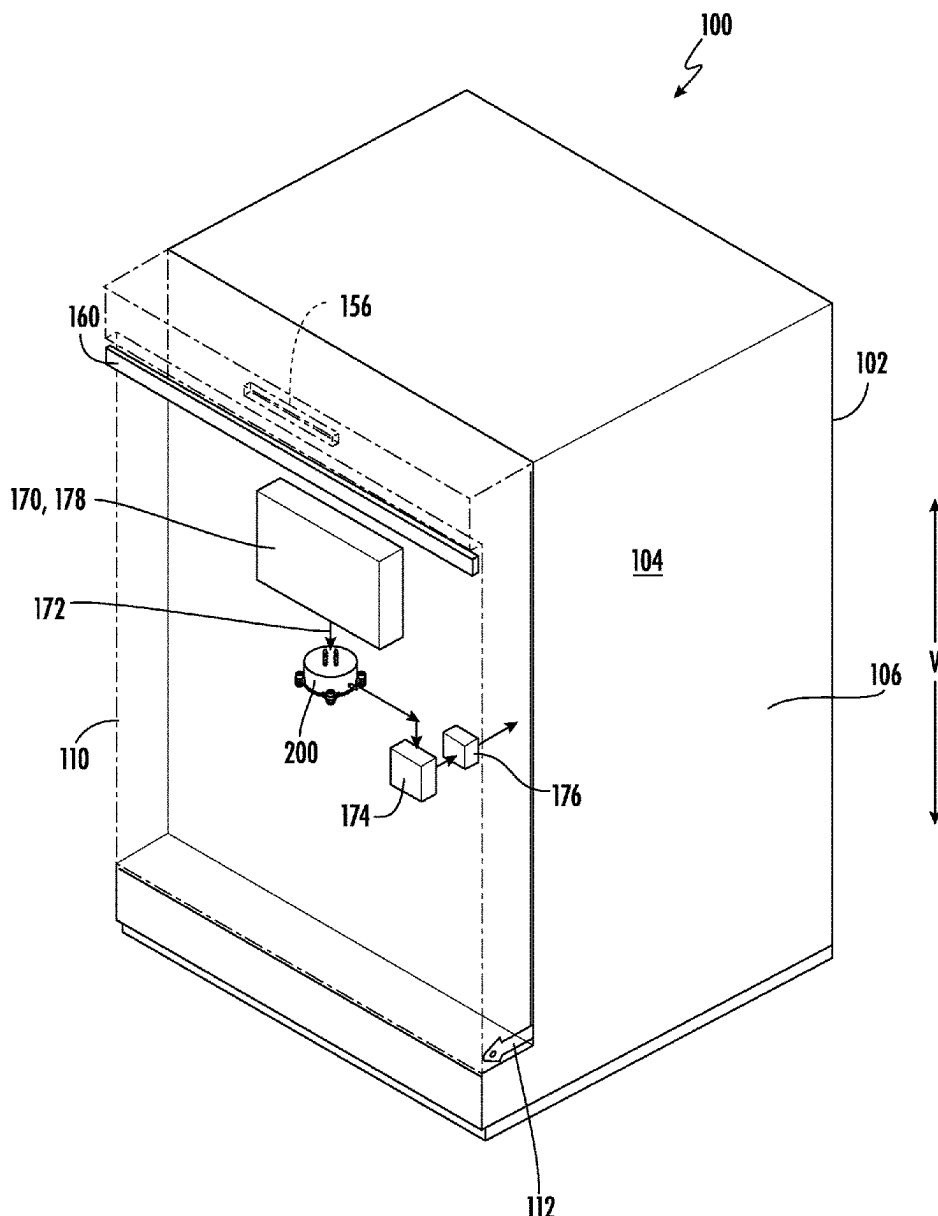
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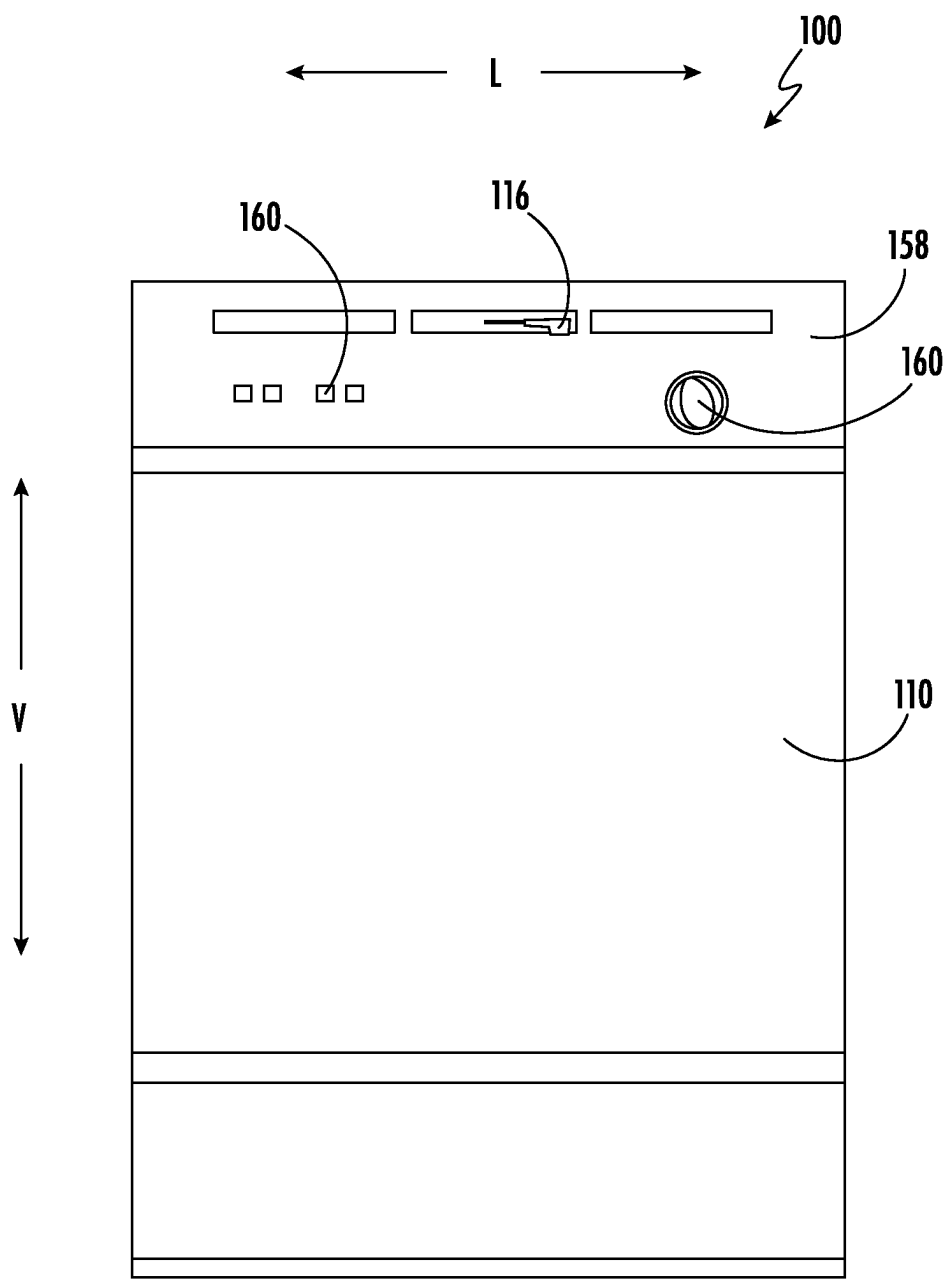
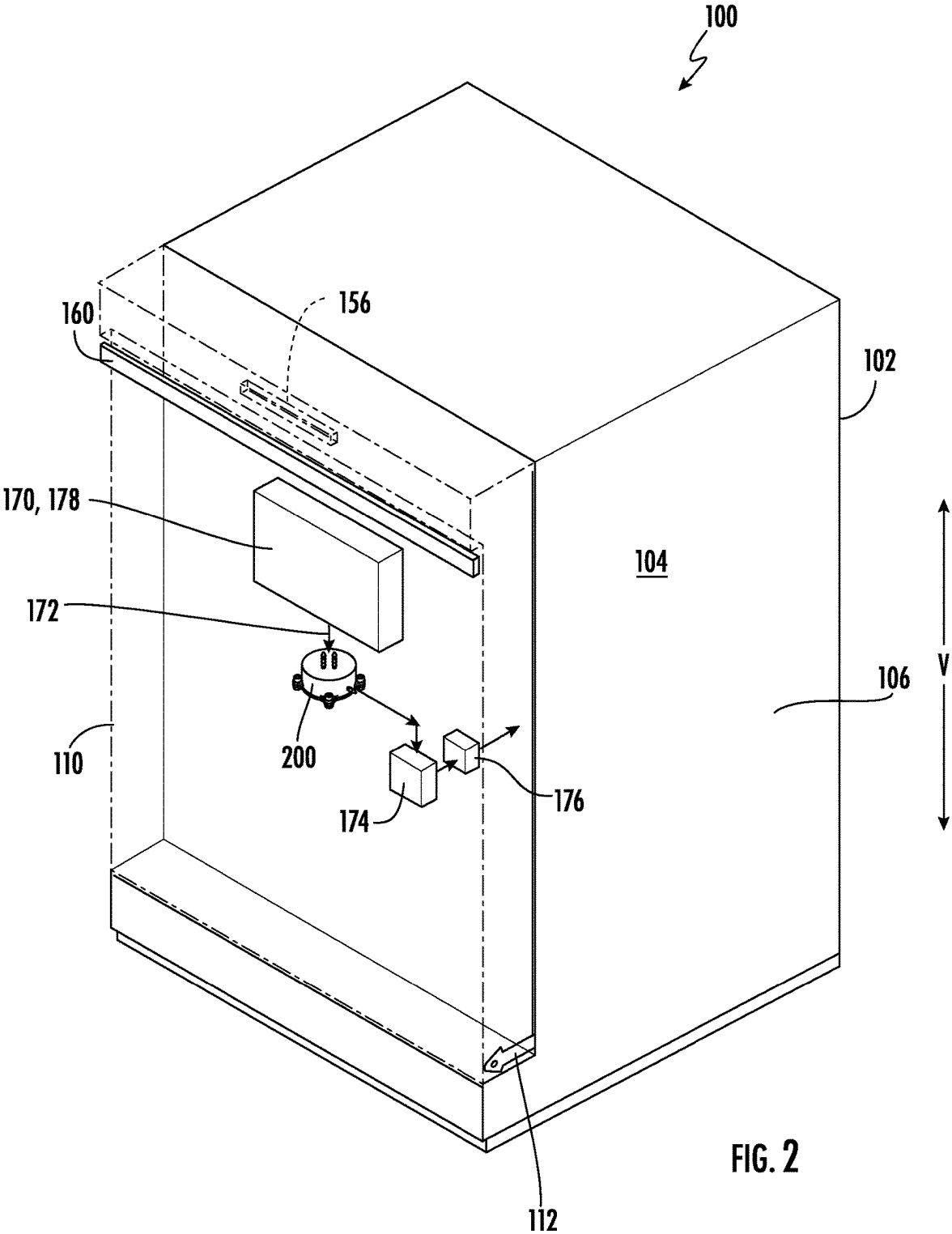


FIG. 1



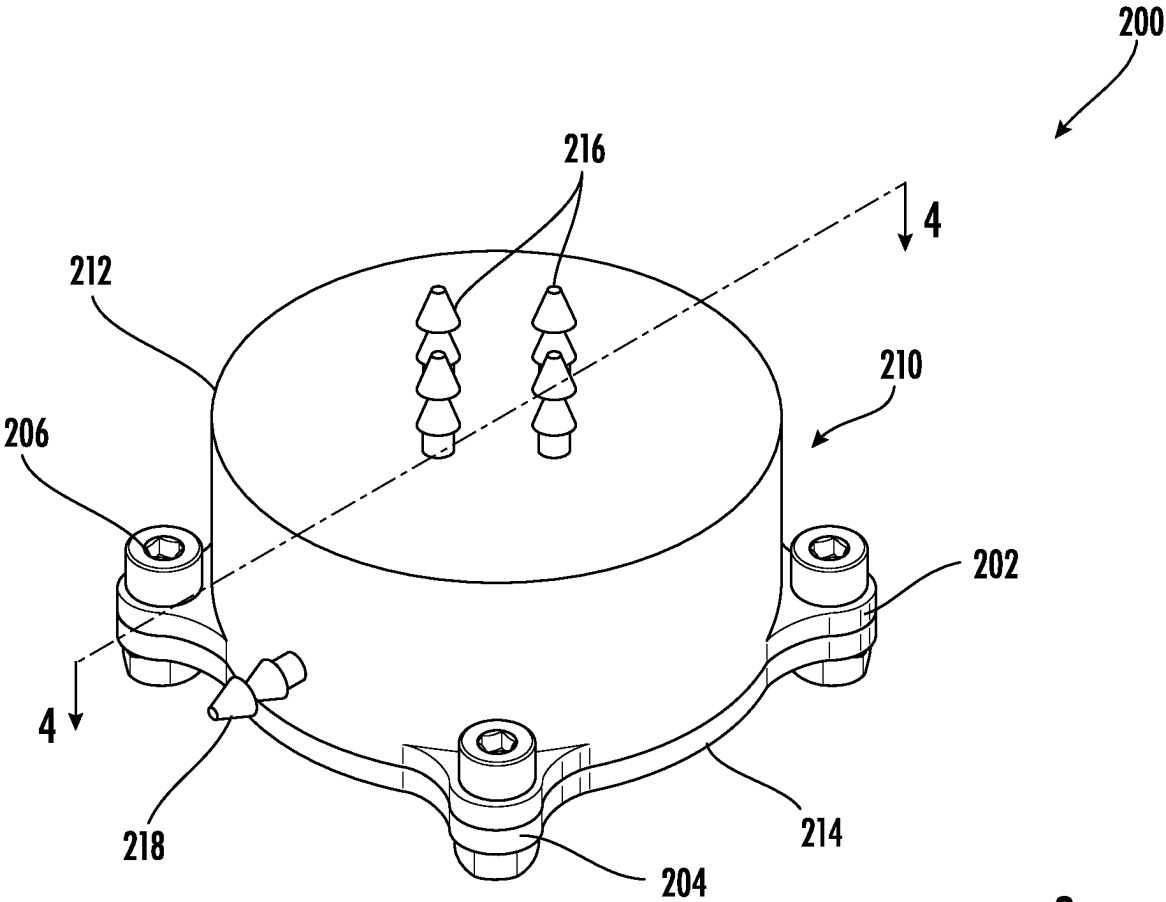


FIG. 3

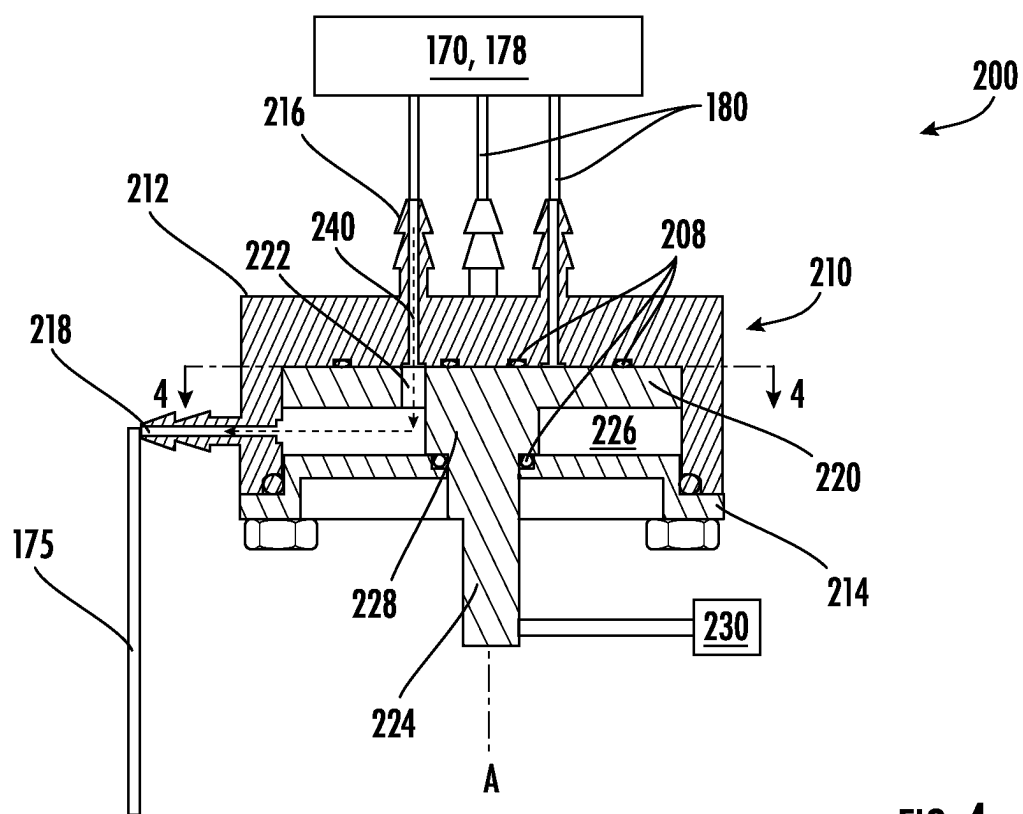


FIG. 4

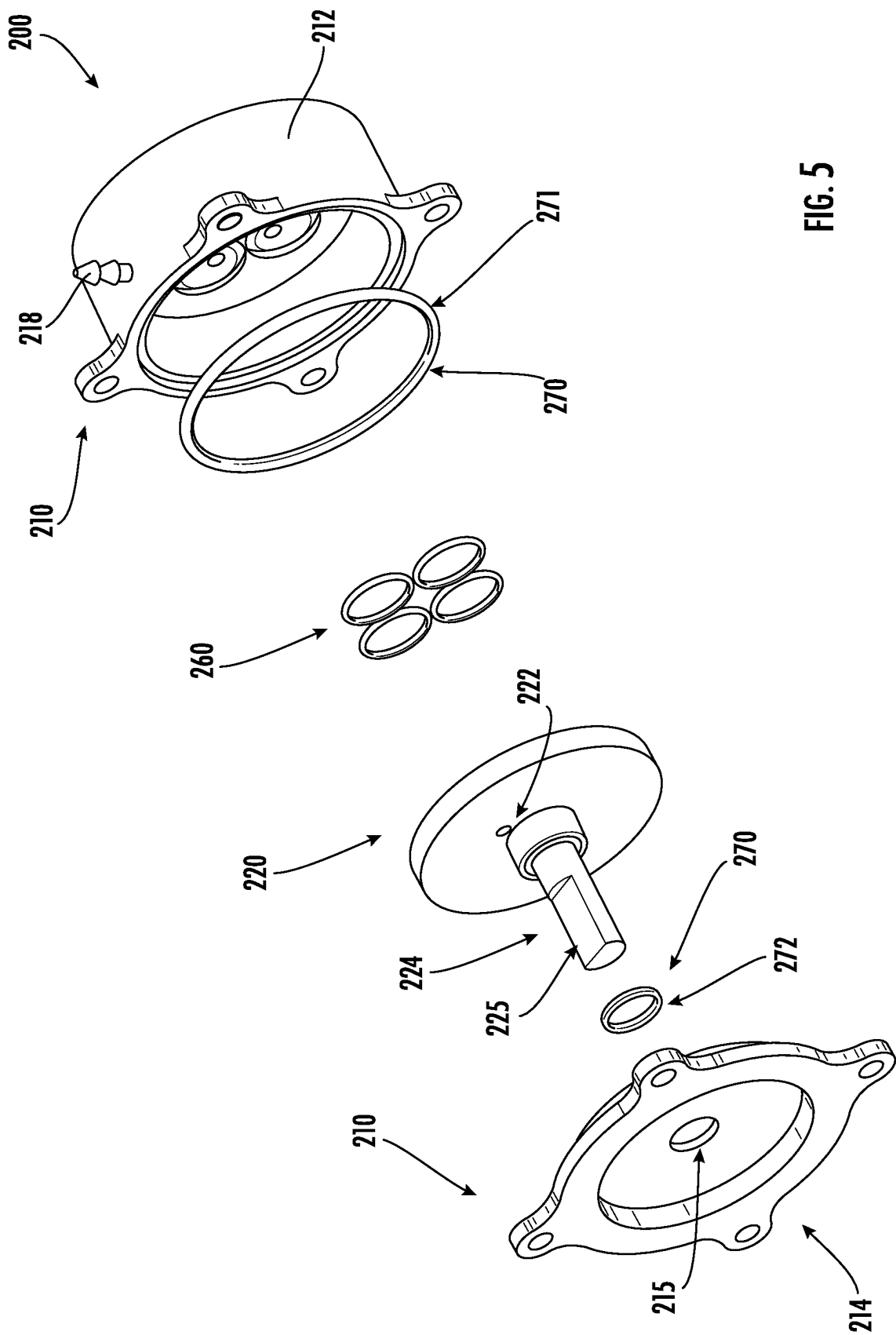
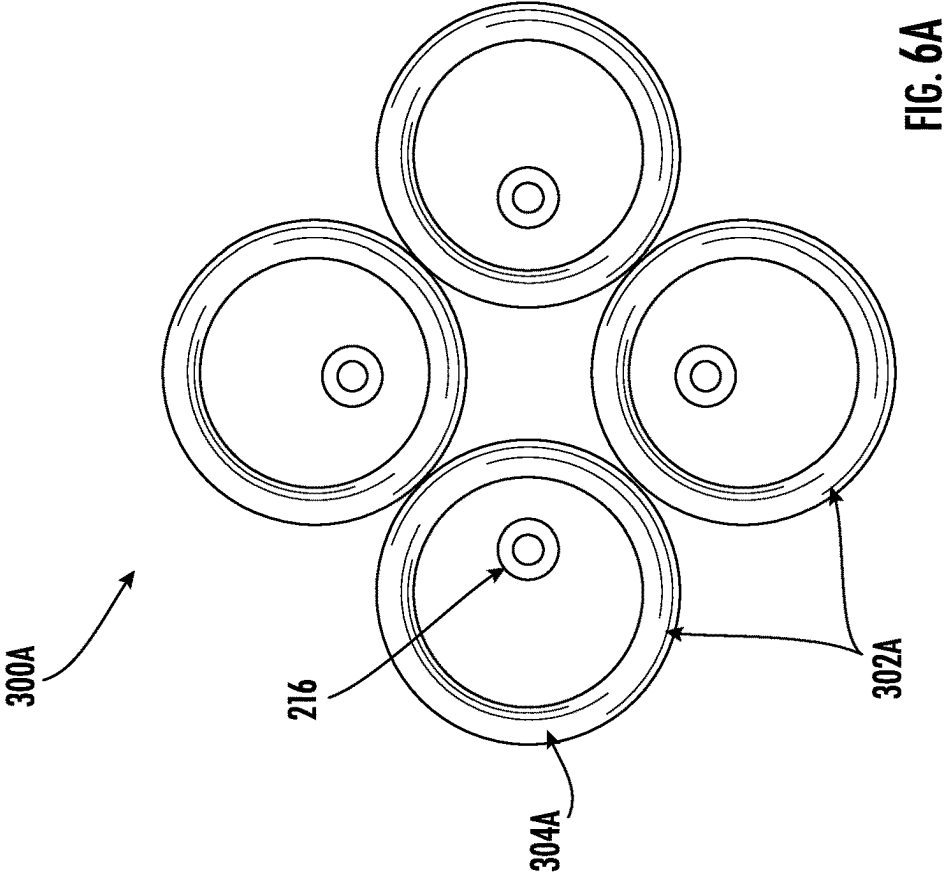
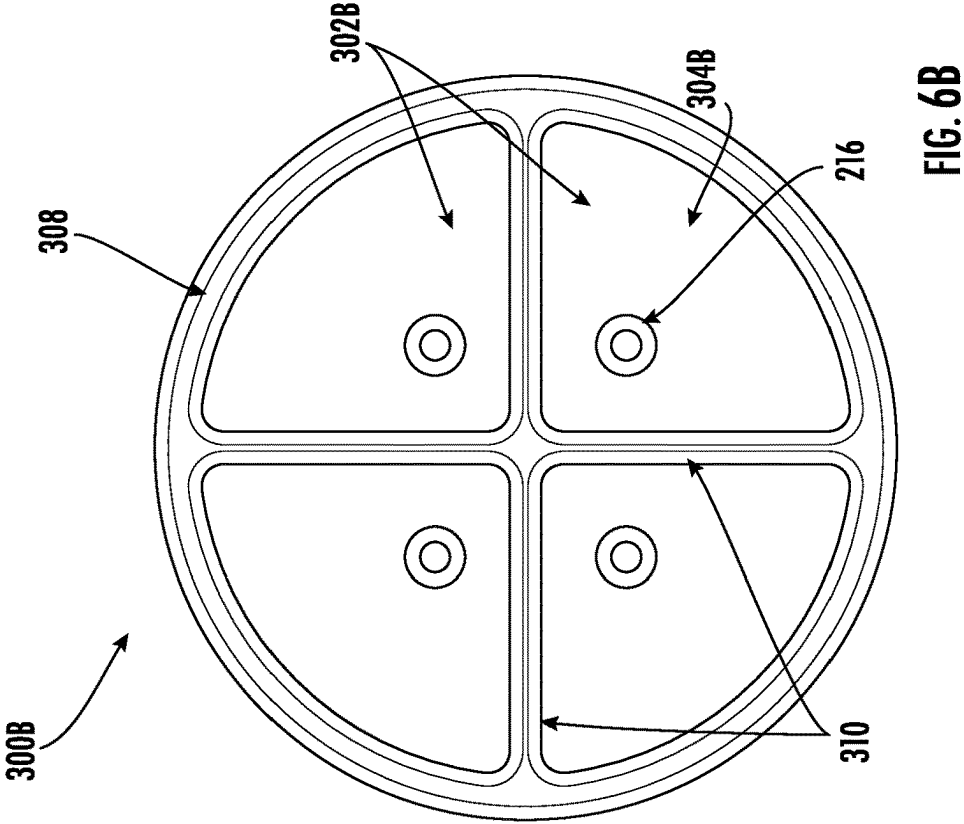


FIG. 5



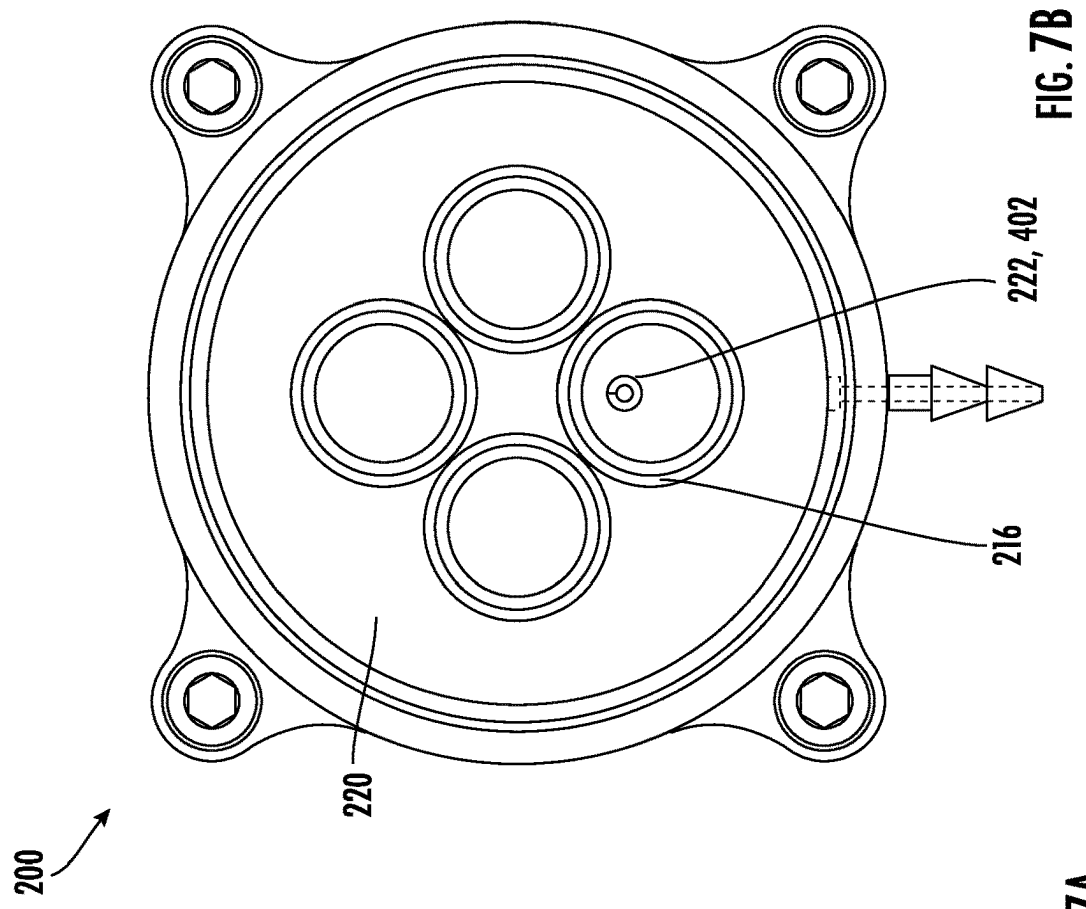


FIG. 7A

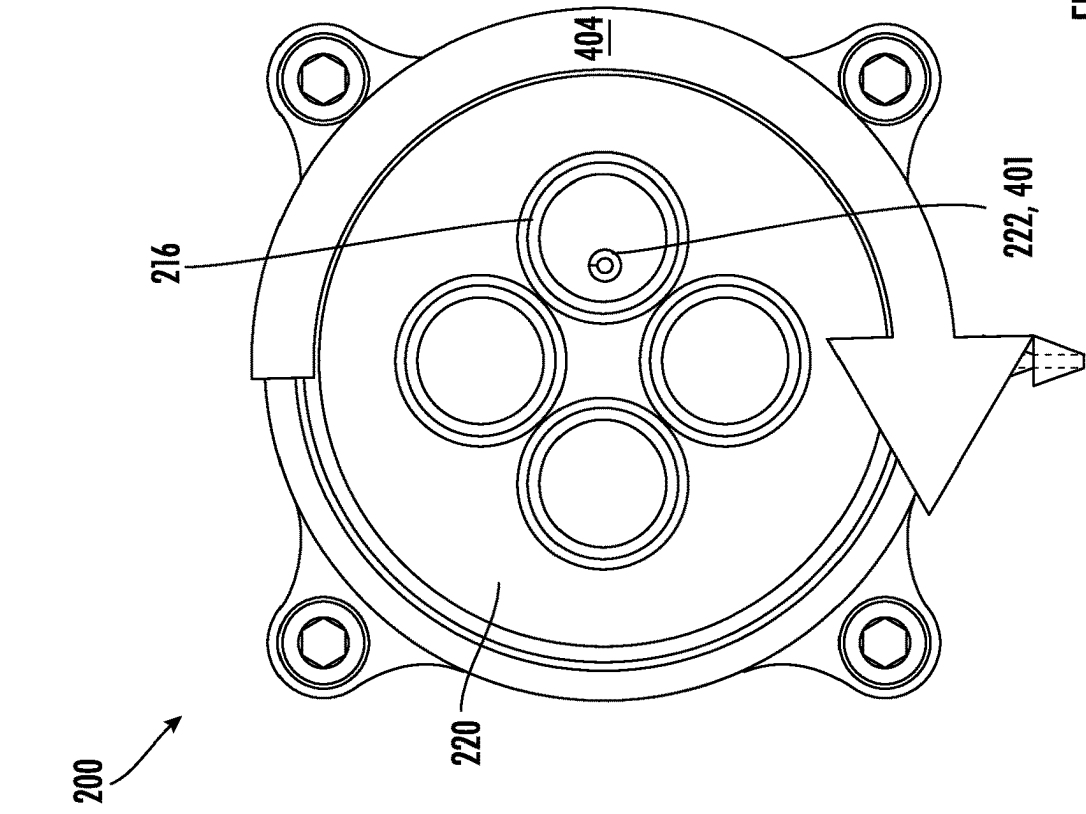


FIG. 7B

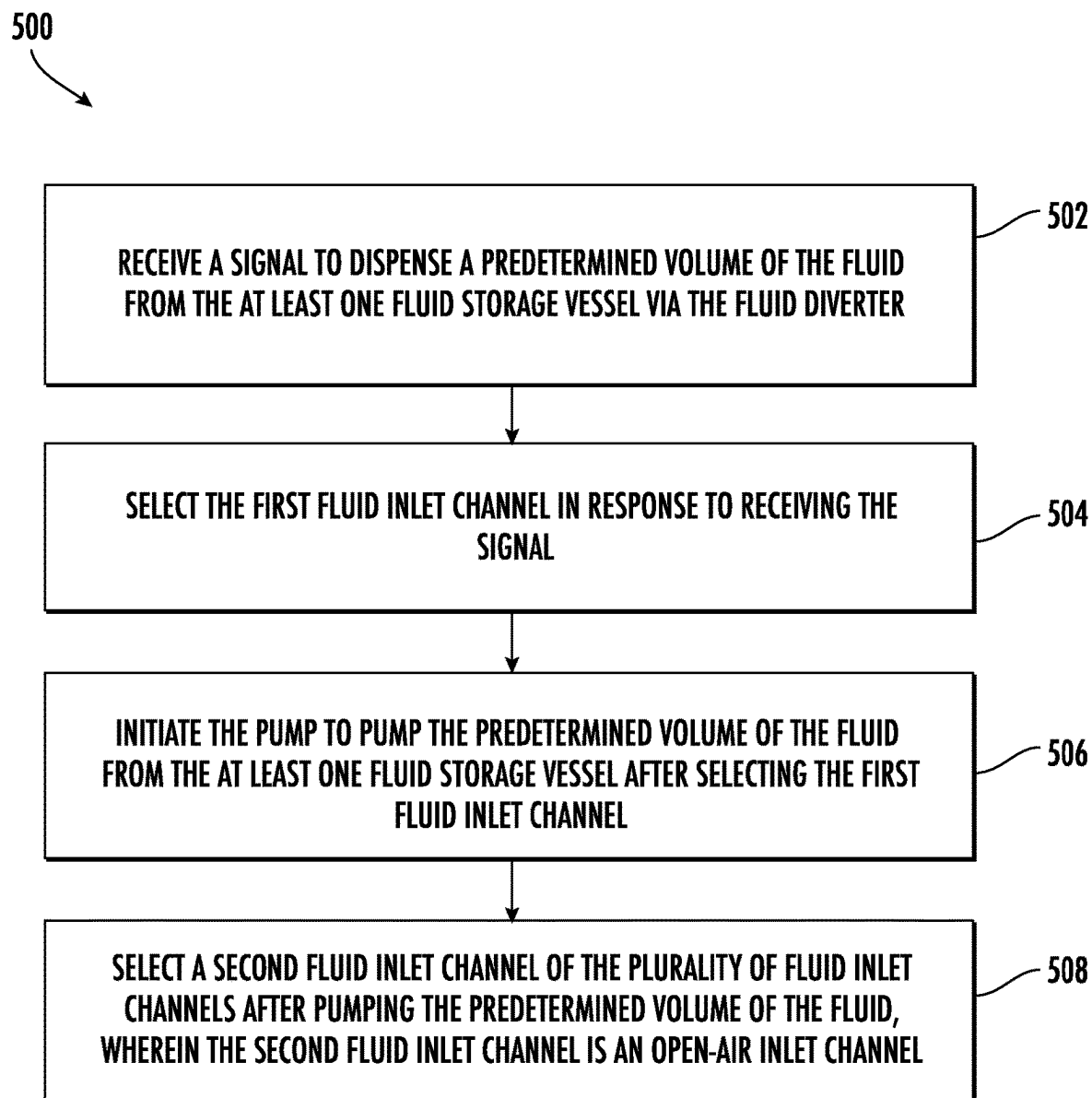


FIG. 8

DISPENSER ASSEMBLY INCLUDING AN OPEN CHANNEL LINE PURGE

FIELD OF THE INVENTION

[0001] The present subject matter relates generally to domestic appliances, and more particularly to dispenser assemblies for dishwasher appliances.

BACKGROUND OF THE INVENTION

[0002] Certain domestic appliances, such as dishwasher appliances, generally include a tub that defines a wash chamber and a pump for directing fluids from a fluid reservoir into the wash chamber during washing operations. A detergent dispenser for dispensing detergent into the wash chamber may be included. In some instances, the detergent dispenser is configured to dispense detergent or other fluids from one or more bulk tanks provided within the appliance. Such dispensers may either mix the fluids or maintain distinct channels for each respective bulk tank, depending on which fluid or detergent is to be added to the wash chamber.

[0003] Mixing the fluids is undesirable in some cases, such a dishwasher cycle where specific liquids need to be dispensed at specific times. Thus, when a first fluid is to be dispensed at a first time and a second fluid is to be dispensed at a second time, the fluids may be mixed within a common flow tube. Further, maintaining distinct channels can be undesirable because separate control devices are required for all channels, thereby adding complexity.

[0004] Accordingly, a dispensing assembly for an appliance that obviates one or more of the above-mentioned drawbacks would be beneficial. In particular, a dispensing assembly including features for purging a common flow tube between dispensations would be useful.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

[0006] In one exemplary aspect of the present disclosure, a dispenser assembly for an appliance is provided. The dispenser assembly may include a fluid diverter including a plurality of fluid inlet channels, a fluid outlet channel, and a diverter input shaft; a pump fluidly coupled to the fluid diverter for selectively urging a flow of fluid from the fluid diverter into the appliance; at least one fluid storage vessel configured to store a fluid, the at least one fluid storage vessel fluidly connected with the fluid diverter via a first fluid inlet channel of the plurality of fluid inlet channels; and a controller operably connected with the fluid diverter and the pump, the controller being configured to perform an operation. The operation may include receiving a signal to dispense a predetermined volume of the fluid from the at least one fluid storage vessel via the fluid diverter; selecting the first fluid inlet channel in response to receiving the signal; initiating the pump to pump the predetermined volume of the fluid from the at least one fluid storage vessel after selecting the first fluid inlet channel; and selecting a second fluid inlet channel of the plurality of fluid inlet channels after pumping the predetermined volume of the fluid, wherein the second fluid inlet channel is an open-air inlet channel.

[0007] In another exemplary aspect of the present disclosure, a method of operating a dispenser assembly of an appliance is provided. The dispenser assembly may include a fluid diverter including a plurality of fluid inlet channels and a fluid outlet channel, a pump fluidly coupled to the fluid diverter, and at least one fluid storage vessel. The method may include receiving a signal to dispense a predetermined volume of a fluid from the at least one fluid storage vessel via the fluid diverter; selecting a first fluid inlet channel in response to receiving the signal, the first fluid inlet channel being fluidly connected with the at least one fluid storage vessel; initiating the pump to pump the predetermined volume of the fluid from the at least one fluid storage vessel after selecting the first fluid inlet channel; and selecting a second fluid inlet channel of the plurality of fluid inlet channels after pumping the predetermined volume of the fluid, wherein the second fluid inlet channel is an open-air inlet channel.

[0008] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

[0010] FIG. 1 provides a front view of a dishwashing appliance in accordance with an exemplary embodiment of the present disclosure.

[0011] FIG. 2 provides a side, cross-sectional view of the dishwashing appliance of FIG. 1.

[0012] FIG. 3 provides a perspective view of a multichannel dispenser assembly in accordance with an exemplary embodiment of the present disclosure.

[0013] FIG. 4 provides a cross sectional view of the multichannel dispenser assembly of FIG. 3 along line 4-4.

[0014] FIG. 5 provides an exploded view of the multichannel dispenser assembly of FIG. 3.

[0015] FIGS. 6A-6B provide alternative embodiments of a seal for a multichannel dispenser assembly in accordance with an exemplary embodiment of the present disclosure.

[0016] FIGS. 7A-7B provide cross-sectional views of the multichannel dispenser assembly of FIG. 3 in different positions.

[0017] FIG. 8 provides a flow diagram of a method for operating a multichannel dispenser assembly in accordance with an exemplary embodiment of the present disclosure.

[0018] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION

[0019] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the

present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0020] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). In addition, here and throughout the specification and claims, range limitations may be combined and/or interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0021] Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “generally,” “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 10 percent margin, i.e., including values within ten percent greater or less than the stated value. In this regard, for example, when used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction, e.g., “generally vertical” includes forming an angle of up to ten degrees in any direction, e.g., clockwise or counterclockwise, with the vertical direction V.

[0022] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” In addition, references to “an embodiment” or “one embodiment” does not necessarily refer to the same embodiment, although it may. Any implementation described herein as “exemplary” or “an embodiment” is not necessarily to be construed as preferred or advantageous over other implementations. Moreover, each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0023] Referring now to the drawings, FIGS. 1 and 2 illustrate a front view of an appliance that may be configured in accordance with aspects of the present disclosure. For the

particular embodiment of FIGS. 1 and 2, the appliance is a dishwasher appliance 100. As shown particularly in FIG. 2, dishwasher appliance 100 may include a cabinet 102 having a tub 104 therein that defines a wash chamber 106 for the receipt of articles for washing. Tub 104 may include a front opening (not shown) and a door 110 for selectively opening and closing wash chamber 106. As shown, door 110 may be hinged to wash chamber 106 for movement between a closed position (shown in FIGS. 1 and 2) and an open position for loading and unloading articles into and from dishwasher 100. In addition, as shown, a latch 116 may be used to lock and unlock door 110 for access to wash chamber 106.

[0024] As shown in FIG. 2, dishwasher appliance 100 may be further equipped with a controller 156 to regulate operation of the dishwasher appliance 100. Controller 156 may include one or more memory devices and one or more microprocessors, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with a cleaning cycle. The memory may represent random access memory such as DRAM or read only memory such as ROM or FLASH. In an embodiment, the processor executes programming instructions stored in the memory. The memory may be a separate component from the processor or may be included onboard within the processor. Further, controller 156 may be positioned in a variety of locations throughout dishwasher appliance 100. In the illustrated embodiment, for example, controller 156 may be located within a control panel area 158 of door 110 as shown in FIG. 1. Typically, controller 156 includes a user interface panel/controls 160 through which a user may select various operational features and modes and monitor progress of dishwasher appliance 100.

[0025] Referring particularly to FIG. 2, dishwasher appliance 100 may further include a fluid reservoir 170 containing a plurality of fluids 178 for washing and/or rinsing the articles within wash chamber 106. Furthermore, as shown, dishwasher appliance 100 may also include a pump 174 for directing or pumping the plurality of fluids 178 from fluid reservoir 170 and into wash chamber 106. Pump 174 may selectively pump fluid (e.g., detergent) from fluid storage vessel 178 (e.g., via a diverter, described below) according to certain wash cycle needs of dishwashing appliance 100. Accordingly, pump 174 may be any suitable pump. In at least one embodiment, pump 174 is a peristaltic pump. However, it should be understood that the described pump is provided by way of example only and the disclosure is not limited to the examples provided herein. Further, and as will be discussed in more detail herein, dishwasher appliance 100 may include a dispenser assembly 200 for controlling the plurality of fluids 178 flowing from fluid reservoir 170 and into wash chamber 106 along a flow path 172.

[0026] As shown in FIG. 2, dispenser assembly 200 may be in fluid communication with a dispenser 176 for dispensing one of the plurality of different fluids into wash chamber 106 (e.g., one different fluid at a time). As shown in FIG. 2, fluid reservoir 170, pump 174, dispenser 176, and dispenser assembly 200 may be disposed within door 110. As such, FIG. 2 depicts fluid reservoir 170, dispenser assembly 200, pump 174, and dispenser 176 as having a particular arrangement along flow path 172. However, it should be understood that such an arrangement is not limiting, and a variety of arrangements can be implemented in accordance with the present disclosure. For example, pump 174 and dispenser

176 may be arranged in a different order along flow path 172 such that dispenser 176 is disposed adjacent or connected to dispenser assembly 200.

[0027] Referring now to FIG. 3, a perspective view of dispenser assembly 200 according to an exemplary embodiment of the present disclosure is illustrated. Dispenser assembly 200 may be used with a variety of appliances, such as dishwasher appliance 100 of FIGS. 1 and 2. As shown, dispenser assembly 200 may include a housing 210 having a plurality of inlets (e.g., fluid inlet channels) 216 and an outlet (e.g., fluid outlet channel) 218. In an embodiment, each of the plurality of inlets 216 may extend from housing 210 in a first direction, whereas outlet 218 may extend from housing 210 in a second direction, the second direction being different from the first direction. As such, in an embodiment, the first and second directions may be generally perpendicular. Further, the plurality of inlets 216 may include any suitable number of inlets, such as four separate inlets as shown. In further embodiments, dispenser assembly 200 may have more or less than four separate inlets.

[0028] Still referring to FIG. 3, housing 210 of dispenser assembly 200 may also have a first housing portion 212 detachably secured to a second housing portion 214. Further, in an embodiment, as shown, first housing portion 212 and second housing portion 214 may be secured together via one or more fasteners 206. In some embodiments, fastener(s) 206 include a variety of different fasteners, such as screws, nuts and bolts, or any other suitable fasteners. In addition, according to an embodiment, first housing portion 212 and second housing portion 214 may each include a plurality of projections 202 and 204, respectively, for securing first housing portion 212 with second housing portion 214. For example, the plurality of projections 202, 204 of first and second housing portions 212, 214 may be aligned and fasteners 206 may extend therethrough.

[0029] Referring now to FIG. 4, a cross-sectional view of dispenser assembly 200 of FIG. 3 along line 4-4 is illustrated. As shown, dispenser assembly 200 may include a rotatable disk 220 positioned within housing 210. Rotatable disk 220 may include an aperture 222. Aperture 222 may be selectively aligned with one of the plurality of inlets 216 of housing 210 to provide a flow path 240 between one of the plurality of inlets 216 and outlet 218.

[0030] A fluid volume 226 may be defined within housing 210. For example, fluid volume 226 may be defined between first housing portion 212 and second housing portion 214. In some embodiments, fluid volume 226 is defined between second housing portion 214 and rotatable disk 220. In such embodiments, rotatable disk 220 includes protrusion 224 extending through second housing portion 214 and beyond housing 210. Rotatable disk 220 may also include a gauge section 228 formed between protrusion 224 and aperture 222 of rotatable disk 220. Gauge section 228 may not extend through or beyond second housing portion 214. In some instances, gauge section 228 is located inside housing 210 and abuts second housing portion 214. Gauge section 228 may separate aperture 222 of rotatable disk 220 from second housing portion 214 such that fluid volume 226 is defined between second housing portion 214 and rotatable disk 220. Thus, as shown in FIG. 4, flow path 240 may pass through fluid volume 226.

[0031] Referring to FIGS. 4 and 5, rotatable disk 220 may include protrusion 224 extending through second housing portion 214 and beyond housing 210. Thus, in some embodi-

ments, protrusion 224 may be mechanically coupled to an actuator 230 for driving rotation of rotatable disk 220 about axis A. Actuator 230 may drive rotatable disk 220 about axis A to align aperture 222 of rotatable disk 220 with one of the plurality of inlets 216. Moreover, protrusion 224 may have any suitable shape, such as a generally cylindrical shape. As shown in FIG. 5, protrusion 224 may have a generally cylindrical shape with a flat surface 225. In such embodiments, flat surface 225 of protrusion 224 may facilitate mechanical coupling with actuator 230 for driving rotation of rotatable disk 220.

[0032] Dispenser assembly 200 may be configured for being fluidly coupled with a fluid reservoir 170 containing a plurality of different fluids 178. Dispenser assembly 200 may be fluidly coupled with fluid reservoir 170 (FIGS. 2 and 4) containing the plurality of fluids 178 via a plurality of conduits 180 coupled to the plurality of inlets 216 of housing 210. Thus, one of the plurality of fluids 178 may flow into fluid volume 226 through one of the plurality of inlets 216 and fill fluid volume 226. Further, outlet 218 of housing 210 may be connected to a pump, such as pump 174 of dishwasher appliance 100 shown in FIG. 1. Accordingly, fluid pumped through dispenser assembly 200 may be provided to dispenser 176 of dishwasher appliance 100 via pump 174. As shown in FIG. 4, a pump line 175 may be connected between outlet 218 of housing 210 and pump 174 of dishwasher appliance 100 for the different fluids 178 to be pumped therethrough. As such, dispenser 176 may be configured to dispense one of the plurality of different fluids 178 into appliance 100 at a time.

[0033] Referring still to FIGS. 4 and 5, dispenser assembly 200 may also include at least one inlet seal 260 within housing 210 to fluidly separate each of the plurality of inlets 216. Dispenser assembly 200 may also include at least one housing seal 270. The housing seal 270 may be arranged between first housing portion 212 and second housing portion 214 and/or protrusion 224 of rotatable disk 220 and second housing portion 214. A first housing seal 271 may be arranged between first and second housing portions 212, 214, and a second housing seal 272 may be arranged between protrusion 224 and second housing portion 214. Further, second housing portion 214 may define an opening 215 for protrusion 224 to extend therethrough.

[0034] Additionally or alternatively, as shown generally in FIGS. 4 and 5, dispenser assembly 200 may include spaces and/or gaps between certain features to facilitate operation thereof. For example, as shown in FIG. 4, a plurality of recesses 208 may be formed into at least one of first housing portion 212 and second housing portion 214. More specifically, one or more of recesses 208 may be formed between first housing portion 212 and rotating disk 220. Additionally or alternatively, one or more of recesses 208 may be formed between first housing portion 212 and second housing portion 214. Furthermore, one of more of recesses 208 may be formed between rotating disk 220 and second housing portion 214. Accordingly, the at least one inlet seal 260 and/or the at least one housing seal 270 may fit within the one or more recesses 208, such as the one or more recesses formed into at least one of first housing portion 212 or second housing portion 214.

[0035] For example, the at least one inlet seal 260 may fit within the one or more recesses 208 formed between first housing portion 212 and rotating disk 220. The at least one housing seal 270 may fit within the one or more recesses 208

formed between first housing portion 212 and second housing portion 214 or the one or more recesses 208 formed between second housing portion 214 and rotating disk 220. First housing seal 271 may fit within recess 208 formed between first and second housing portions 212, 214, while second housing seal 272 may fit within recess 208 between second housing portion 214 and rotating disk 220. In some embodiments, as shown in FIG. 5, inlet seal(s) 260 and/or housing seal(s) 270 may include one or more O-rings.

[0036] Referring to FIGS. 6A-6B, further embodiments of inlet seal(s) 360 of dispenser assembly 200 are illustrated according to the present disclosure. For instance, FIG. 6A provides a multi-seal arrangement 300A of inlet seal 260. As shown, multi-seal arrangement 300A may include a plurality of inlet seals 302A with one of the plurality of inlet seals 302A being arranged with each of the plurality of inlets 216. The plurality of inlet seals 302A may include a first inlet seal 304A arranged with one of the plurality of inlets 216. Accordingly, the plurality of inlet seals 302A may include a number of inlet seals that is equivalent to a number of each inlet of the plurality of inlets 216.

[0037] FIG. 6B illustrates a segmented-ring seal 300B of inlet seal 260. Segmented-ring seal 300B may include a single inlet seal that defines a plurality of openings 302B. One of the plurality of openings 302B may be arranged with each of the plurality of inlets 216. As shown in FIG. 6B, the second inlet seal includes a ring 408 and a plurality of cross-connecting segments 410, thereby defining the plurality of openings 302B. The plurality of openings 302B may include a first opening 304B arranged with one of the plurality of inlets 216. Accordingly, the plurality of openings 302B may include a number of openings that is equivalent to a number of inlets of the plurality of inlets 216 (e.g., four).

[0038] Referring now to FIGS. 7A-7B, cross-sectional views of different positions of the dispenser assembly 200 of FIG. 3 are illustrated according to the present disclosure. The cross-sectional views depicted in FIGS. 7A-7B are described herein with further reference to FIGS. 1-5 and 6A-6B. Further, in FIGS. 7A-7B, the cross-sectional views are taken along the line 4-4 shown in FIG. 4. As such, dispenser assembly 200 may include rotatable disk 220 with aperture 222 and inlet seal 260 disposed between rotatable disk 220 and first housing portion 212. Referring particularly to FIG. 7A, aperture 222 of rotatable disk 220 may be in a first position 401. In the first position 401, aperture 222 is selectively aligned with an inlet (e.g., a first inlet or first fluid inlet channel) of the plurality of inlets 216 to provide a flow path 240 between the first fluid inlet channel of the plurality of inlets 216 and outlet 218 of housing 210. In the first position 401, the remaining inlets of the plurality of inlets 216 are blocked by rotatable disk 220 while allowing the first fluid inlet channel to facilitate releasing one of the plurality of fluids 178 contained within fluid reservoir 170. Actuator 230 of dispenser assembly 200 may be configured to drive rotation of rotatable disk 220 about axis A. As such, rotatable disk 220 may be rotated in a rotational direction 404 such that aperture 222 is no longer in the first position 401.

[0039] As shown in FIG. 7B, rotatable disk 220 may be rotated such that aperture 222 is in a second position 402. In the second position 402, aperture 222 is selectively aligned with a different inlet (e.g., a second inlet or second fluid inlet channel) of the plurality of inlets 216 to provide a flow path 240 between the second fluid inlet channel of the plurality of

inlets 216 and outlet 218 of housing 210. As will be described in more detail below, the second fluid inlet channel may be an open-air inlet. In detail, no fluid from the plurality of fluids 178 from fluid reservoir 170 is connected to the second fluid inlet channel. Furthermore, it should be understood that any of the plurality of inlets 216 may be referred to as the second fluid inlet channel or open-air channel. Thus, any of the plurality of inlets 216 may be an open-air inlet channel. When rotatable disk 220 is in the second position 402, air (e.g., ambient air) may be pumped through the second fluid inlet channel and into, e.g., flow path 172, pump line 175, etc. As such, flow path 172 may be purged of any residual fluid remaining in flow path 172, pump line 172, or the like.

[0040] Now that the general descriptions of an exemplary appliance have been described in detail, a method 500 of operating an appliance (e.g., dishwasher appliance 100) will be described in detail. Although the discussion below refers to the exemplary method 500 of operating dishwasher appliance 100, one skilled in the art will appreciate that the exemplary method 500 is applicable to any suitable domestic appliance capable of performing a dispensing operation (e.g., such as a washing machine appliance, etc.). In exemplary embodiments, the various method steps as disclosed herein may be performed by controller 156 and/or a separate, dedicated controller. FIG. 8 provides a flow chart illustrating a method of operating a dishwasher appliance. Hereinafter, method 500 will be described with specific reference to FIG. 8.

[0041] At step 502, method 500 may include receiving a signal to dispense a predetermined volume or amount of fluid from at least one fluid storage vessel via a fluid diverter. As discussed above, fluid storage vessels or bulk tanks may be included within or fluidly attached to the appliance (e.g., dishwasher appliance 100). Further, the dispenser assembly (e.g., dispenser assembly 200) may be in fluid communication with each of the fluid storage vessels as well as a chamber of the appliance. The signal may include a user input to initiate a washing operation, such as a dishwashing operation. Accordingly, the signal may be received via a user interface of the appliance, via a remote input from a remote device, via an automatic or preprogrammed request, or the like.

[0042] At step 504, method 500 may include selecting a first fluid inlet channel in response to receiving the signal. The first fluid inlet channel may be one of a plurality of fluid inlet channels (e.g., inlets 218). The first fluid inlet channel may be in fluid communication with a first bulk storage vessel or tank, such as a detergent tank. In selecting the first fluid inlet channel, method 500 may include rotating, via an actuator (e.g., motor), a rotatable disk (e.g., rotatable disk 220) to a first position. As discussed above, the first position may align an aperture (e.g., through hole or aperture 222) with the first fluid inlet channel. The signal may thus include a command to operate the actuator (or motor) to rotate the rotatable disk to a predetermined position (e.g., first position) or through a predetermined rotational arc.

[0043] At step 506, method 500 may include initiating a pump to pump the predetermined volume of fluid from the at least one fluid storage vessel. In detail, after selecting the first fluid inlet channel, method 500 may direct the pump (e.g., pump 174) at a predetermined level to supply the fluid from the at least one fluid storage vessel to the fluid diverter and subsequently to the wash chamber of the appliance. The

signal to dispense the predetermined volume of fluid may include the predetermined volume of fluid, for instance, in milliliters, ounces, or the like. For instance, according to a selected operation (e.g., cleaning cycle), the pump may operate according to a predetermined length of time to accurately dispense the predetermined volume of fluid. In additional or alternative embodiments, the pump may be operated according to a predetermined pattern, such as a number of rotations, iterations, cycles, or the like (e.g., in the case of a peristaltic pump). Accordingly, the predetermined pattern or length of time may be based on the predetermined volume of fluid. It should be understood that any suitable pump may be incorporated and any suitable measurement of the fluid may be used in specific embodiments.

[0044] At step **508**, method **500** may include selecting a second fluid inlet channel of the plurality of fluid inlet channels after pumping the predetermined volume of fluid. The second fluid inlet channel may be an open-air inlet channel. As described above, the plurality of fluid inlet channels may include at least one fluid inlet channel (e.g., the second fluid inlet channel) that is not connected to any tank, vessel, or storage container. As such, air (e.g., ambient air) may be pumped through the second fluid inlet channel.

[0045] In selecting the second fluid inlet channel, method **500** may include rotating, via the actuator (e.g., motor), the rotatable disk to a second position. As discussed above, the second position may align the aperture (e.g., through hole or aperture **222**) with the second fluid inlet channel (e.g., the open-air inlet channel). The signal may thus include a command to operate the actuator (or motor) to rotate the rotatable disk to a predetermined position (e.g., second position) or through a predetermined rotational arc.

[0046] Method **500** may further include initiating the pump to purge the fluid from the pump line after selecting the second fluid inlet channel. For instance, after aligning the aperture or through hole with the second fluid inlet channel (e.g., the open-air channel), the pump may be activated to draw air through the second fluid inlet channel and into the fluid diverter. As mentioned above, one or more flow lines, such as the pump line, may be provided downstream from the fluid diverter. The one or more flow lines may further include an inlet line into the wash chamber of the appliance. By operating the pump after selecting the open-air channel, residual fluid within the flow lines (e.g., the pump line, the inlet line, etc.) may be purged or flushed into the wash chamber. Advantageously, method **500** may ensure that an entire determined dosage of the fluid (e.g., detergent) is supplied to the wash chamber. Additionally or alternatively, method **500** may ensure that two or more fluids are not inadvertently mixed within the flow lines by purging the flow lines after supplying a fluid therethrough.

[0047] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

What is claimed is:

1. A dispenser assembly for an appliance, the dispenser assembly comprising:
 - a fluid diverter comprising a plurality of fluid inlet channels, a fluid outlet channel, and a diverter input shaft;
 - a pump fluidly coupled to the fluid diverter for selectively urging a flow of fluid from the fluid diverter into the appliance;
 - at least one fluid storage vessel configured to store a fluid, the at least one fluid storage vessel fluidly connected with the fluid diverter via a first fluid inlet channel of the plurality of fluid inlet channels; and
 - a controller operably connected with the fluid diverter and the pump, the controller being configured to perform an operation, the operation comprising:
 - receiving a signal to dispense a predetermined volume of the fluid from the at least one fluid storage vessel via the fluid diverter;
 - selecting the first fluid inlet channel in response to receiving the signal;
 - initiating the pump to pump the predetermined volume of the fluid from the at least one fluid storage vessel after selecting the first fluid inlet channel; and
 - selecting a second fluid inlet channel of the plurality of fluid inlet channels after pumping the predetermined volume of the fluid, wherein the second fluid inlet channel is an open-air inlet channel.
2. The dispenser assembly of claim 1, wherein the fluid diverter further comprises:
 - a housing, wherein each of the plurality of fluid inlet channels and the fluid outlet channel is formed through the housing;
 - a rotatable disk positioned within the housing, the rotatable disk comprising an aperture for being selectively aligned with one of the plurality of fluid inlet channels to provide a flow path between one of the plurality of fluid inlet channels and the fluid outlet channel; and
 - an actuator mechanically coupled to the rotatable disk for driving rotation of the rotatable disk about an axis to align the aperture of the rotatable disk with one of the plurality of fluid inlet channels.
3. The dispenser assembly of claim 2, wherein selecting the first fluid inlet channel comprises:
 - rotating, via the actuator, the rotatable disk to a first position at which the aperture is aligned with the first fluid inlet channel.
4. The dispenser assembly of claim 3, wherein selecting the second fluid inlet channel comprises:
 - rotating, via the actuator, the rotatable disk to a second position at which the aperture is aligned with the open-air inlet channel.
5. The dispenser assembly of claim 1, further comprising:
 - a pump line fluidly coupled between the fluid outlet channel and the pump.
6. The dispenser assembly of claim 5, wherein the operation further comprises:
 - initiating the pump to purge the fluid from the pump line after selecting the second fluid inlet channel.
7. The dispenser assembly of claim 1, wherein initiating the pump to pump the predetermined volume of fluid from the at least one fluid storage vessel comprises:
 - activating the pump according to a predetermined pattern.
8. The dispenser assembly of claim 7, wherein the predetermined pattern is based on the predetermined volume of the fluid.

9. The dispenser assembly of claim 1, wherein the appliance is a dishwasher appliance comprising a tub and a door moveable between an open position and a closed position, wherein the dispenser assembly is provided within the door.

10. A method of operating a dispenser assembly of an appliance, the dispenser assembly comprising a fluid diverter comprising a plurality of fluid inlet channels and a fluid outlet channel, a pump fluidly coupled to the fluid diverter, and at least one fluid storage vessel, the method comprising:

receiving a signal to dispense a predetermined volume of a fluid from the at least one fluid storage vessel via the fluid diverter;

selecting a first fluid inlet channel in response to receiving the signal, the first fluid inlet channel being fluidly connected with the at least one fluid storage vessel;

initiating the pump to pump the predetermined volume of the fluid from the at least one fluid storage vessel after selecting the first fluid inlet channel; and

selecting a second fluid inlet channel of the plurality of fluid inlet channels after pumping the predetermined volume of the fluid, wherein the second fluid inlet channel is an open-air inlet channel.

11. The method of claim 10, wherein the fluid diverter further comprises:

a housing, wherein each of the plurality of fluid inlet channels and the fluid outlet channel is formed through the housing;

a rotatable disk positioned within the housing, the rotatable disk comprising an aperture for being selectively aligned with one of the plurality of fluid inlet channels to provide a flow path between one of the plurality of fluid inlet channels and the fluid outlet channel; and

an actuator mechanically coupled to the rotatable disk for driving rotation of the rotatable disk about an axis to align the aperture of the rotatable disk with one of the plurality of fluid inlet channels.

12. The method of claim 11, wherein selecting the first fluid inlet channel comprises:

rotating, via the actuator, the rotatable disk to a first position at which the aperture is aligned with the first fluid inlet channel.

13. The method of claim 12, wherein selecting the second fluid inlet channel comprises:

rotating, via the actuator, the rotatable disk to a second position at which the aperture is aligned with the open-air inlet channel.

14. The method of claim 10, further comprising:

a pump line fluidly coupled between the fluid outlet channel and the pump.

15. The method of claim 14, further comprising:

initiating the pump to purge the fluid from the pump line after selecting the second fluid inlet channel.

16. The method of claim 10, wherein initiating the pump to pump the predetermined volume of fluid from the at least one fluid storage vessel comprises:

activating the pump according to a predetermined pattern.

17. The method of claim 16, wherein the predetermined pattern is based on the predetermined volume of the fluid.

18. The method of claim 10, wherein the appliance is a dishwasher appliance comprising a tub and a door moveable between an open position and a closed position, wherein the dispenser assembly is provided within the door.

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